

Reviewer Pack — Persistent Homology of Read-Twice BPs Bounded by Log-Size

Entry b4f04707e5bb · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Persistent Homology of Read-Twice BPs Bounded by Log-Size

Entry ID: b4f04707e5bb **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-12 08:14:55 UTC

1. Conjecture

Field A (mathematical branch): Persistent Homology **Field B** (complexity object): Read-Twice Branching Programs

Statement:

For a read-twice BP P computing a function $f: \{0,1\}^n \rightarrow \{0,1\}$, let $\beta(P) = \sum_k \{\dim H_k\} \cdot \text{persistence}_k$, where H_k is the k -th homology group of P 's computation graph. Then $\beta(P) = O(\log \text{size}(P))$, but $\beta(\text{IP}_2) = \Omega(n^2)$ where IP_2 is the inner product mod 2 function.

Rationale (proposer's reasoning):

Persistent homology captures hierarchical topological features of computation graphs. Read-once BPs have acyclic structures limiting homology complexity, while read-twice BPs may encode richer topological patterns. IP_2 's exponential BP complexity creates persistent high-dimensional holes.

Taxonomy category: SOS_HIERARCHY (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): cc8b8ce10bc52b53

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

default: $\geq 80\%$ seeds must support, no counterexample

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.00	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.95	UNCERTAIN	SAFE

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
KARP_LIPTON	SAFE	0.95	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=0, elapsed=1.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_read_twice_bp(n):
        bp = []
        for _ in range(n):
            bp.append(random.choice([0, 1]))
            bp.append(random.choice([0, 1]))
        return bp

    def compute_persistence(bp):
        # Simplified persistence computation (not actual GUDHI)
        homology_groups = [len(set(bp[:i])) for i in range(1, len(bp) + 1)]
        persistence_values = [homology_groups[i] - homology_groups[i-1] for i in range(1, len(homology_groups))]
        return sum(persistence_values)

    def size(bp):
        return len(bp)

    n = 40
    bp = generate_read_twice_bp(n)
    beta_p = compute_persistence(bp)
    log_size_p = math.log2(size(bp))

    if beta_p > log_size_p:
        conjecture_holds = False
        counterexample = "beta(P) exceeds 0(log size(P))"
    else:
        conjecture_holds = True
        counterexample = ""

    return {
        "metric_name": "beta(P)",
        "metric_value": beta_p,
        "instances_tested": 1,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }
```


gregate_stats and per_seed_brief are empty, making it impossible to assess variability or detect adversarial cases.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

All 10 trials used the same single instance, violating the n-too-small failure mode. The support_fraction=1.0 is statistically vacuous for scaling claims. | next: Test with ≥ 80 diverse instances across varying n to validate scaling behavior

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	40178
2	propose	ollama_remote	qwen3:8b	0	0	40957
3	preregistration	ollama_remote	qwen3:8b	0	0	24183
4	novelty	ollama_remote	qwen3:8b	0	0	27092
5	novelty	ollama_remote	qwen3:8b	0	0	17359
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10679
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	39697
8	critic	ollama_remote	qwen3:8b	0	0	134445
9	judge	ollama_remote	qwen3:8b	0	0	17357

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 351947 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in research/audit/b4f04707e5bb.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/b4f04707e5bb.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/b4f04707e5bb.tar.gz (if generated)